

DeploySlot

Product Requirement Document

1. High-Level Business Context & Objective

Target Audience

Release Managers, Engineering Managers (EMs), Product Owners (POs), QA Manager and a triage squad consisting of a Project Manager (PM), TL, QA, Shared DevOps, and an SRE.

The Problem

In an enterprise with 200+ technical members, teams operate in isolated project silos with zero cross-project visibility into sprints or planned release dates. Currently, all slot booking happens manually through chaotic email trails, phone calls, and instant messages. The Release Manager must manually check spreadsheets to prevent collisions. This manual process causes scheduling bottlenecks, human error, double-booked shared resources (DevOps/SRE), and delayed deployments.

The MVP Solution

A streamlined **Self-Service Release Slot Booking** Portal or Mobile App. This focused MVP provides a single source of truth for scheduling. It replaces emails and phone calls with a self-service calendar grid that automatically blocks slot collisions, checks the availability of shared resources, enforces a mandatory PM readiness checklist, and handles an automated 5-stage digital approval chain right inside the application.

2. System Structure & Simplified Resource Matrix

The application handles a flat multi-project list while maintaining blind slot status visibility to respect project privacy.

Resource Mapping

- Dedicated Team: Project Manager (PM), Technical Lead, QA Lead.
- Shared Enterprise Pool: Shared DevOps Engineer, Site Reliability Engineer (SRE).

3. Core Modules & Feature Requirements

Module 1: Automated Self-Service Slot Engine

- Fixed Time Slots: The calendar divides the day into distinct, non-overlapping 2-hour windows based on India Business Hours.

- Blind Availability Status: When a user views the calendar, slots booked by other project silos simply display as "Unavailable."
- Automated Collision Block: The backend prevents any two projects from successfully booking the exact same time slot window.

Module 2: Shared Resource Guardrail

The portal blocks form submission unless a unique engineer is assigned to the shared roles.

- The system evaluates the shared enterprise database. If DevOps_Engineer_A or SRE_A is already assigned to an active slot during that 2-hour window, the portal marks them as "Unavailable" for any other project attempting a booking.

Module 3: PM Pre-Scheduling Checklist Gatekeeper

Before a Project Manager can submit the form and route it to the approval queue, they must explicitly interact with a Mandatory Readiness Checklist inside the wizard interface:

- Release Notes Upload: PM must upload a predefined template detailing user-facing changes and bug fixes.
- Deployment Steps: PM must upload a predefined template for Devops to execute during deployment.
- Handover to DevOps/SRE: PM must upload the technical deployment runbook document containing rollback paths and alert thresholds.
- Enforcement: The "Submit Request" button remains completely disabled until both files are uploaded and both checkboxes are checked.

Module 4: 5-Stage Sequential Approval Workflow

Once the PM submits the checklist-validated request, it enters a structured status lifecycle. The release cannot jump stages; it must pass through each sign-off sequentially:

[Pending QA Approval] —► [Pending EM Approval] —► [Pending Product Approval] —► [Pending DevOps/SRE] —► [Pending Release Mgr] —► [Approved & Scheduled]

1. Stage 1: QA Signoff: The dedicated QA Manager reviews test logs and clicks "Approve" -> moves to Pending EM Approval.
2. Stage 2: EM Signoff: The project's Engineering Manager reviews code metrics and clicks "Approve" -> moves to Pending Product Approval.
3. Stage 3: Product/Business Signoff: The Product Owner checks business readiness and clicks "Approve" -> moves to Pending DevOps/SRE Approval.
4. Stage 4: DevOps/SRE Signoff: The shared infrastructure team verifies the runbook environment metrics and clicks "Approve" -> moves to Pending Release Manager Approval.

5. Stage 5: Release Manager Final Approval: The master Release Manager reviews global environment sanity. Clicking "Approve" transitions the window to its final execution state: Approved & Scheduled.

4. Step-by-Step Release Scheduling Wizard

[Step 1: Details & Resources] —► [Step 2: Mandatory PM Checklist & Uploads] —► [Step 3: Auto-Conflict Check & Route]

- Step 1: Parameters: The PM selects their Project ID, picks an available 2-hour time block on the calendar, and assigns names to the mandatory squad fields (PM, Dev, QA, Shared DevOps, SRE).
- Step 2: PM Checklist: The PM uploads the Release Notes and Handover Documents, then ticks both confirmation checkboxes to unlock the form.
- Step 3: Submission & Validation: The system evaluates the database transaction. If clear, it locks the resource schedules and routes the release directly into the QA Signoff Queue (Stage 1).

5. MVP Critical Engineering Rules

1. The Slot Collision & Overlap Rule

The booking logic must enforce absolute mathematical isolation on time blocks. If an overlap occurs during a concurrent write request for the same environment target, the transaction must abort and roll back to protect calendar integrity.

2. Double-Booking Constraint

The verification engine treats shared resources globally. If an SRE or DevOps engineer is committed to Project X from 7:00 AM to 9:00 AM IST, any parallel submission from another project attempting to claim their time within those exact hours must fail validation.

3. State Invariance & Immutability

- Order Enforcement: The backend database must reject an approval update if the preceding signature in the 5-stage matrix is missing or out of sequence.
- Locking: Once a slot transitions to "Approved & Scheduled", all fields become strictly read-only. If any stage clicks "Reject", the user must click "Cancel Release," which deletes the entry, frees up the slot globally, and resets the process back to the open pool.

3. Admin custom Blackout Overrides

The master Release Manager (Global Admin) must have the ability to explicitly lock and flag specific days, specific hour blocks, or entire weekends as "**Unavailable**" directly from an admin control panel. This override must instantly block normal project scheduling for scenarios such as

region-specific business hour boundaries, heavy system load windows, special corporate occasions, **"No-Deploy Fridays"** (to prevent weekend incidents), and long festive holidays. When an admin blocks these windows, they must immediately show as "Unavailable" across all siloed project views.

3. Automated Stakeholder Confirmation Notifications:

Upon final Stage 5 approval by the Release Manager, the system must instantly trigger a comprehensive confirmation email to all involved stakeholders. This email serves as the official deployment pass and must include all vital execution details: the confirmed Date and Time Slot (IST), Project ID, Build Version, assigned Triage Squad names, and direct links to the view Release Notes and SRE Handover runbook.

Future Scope — Phase 2 Blueprint

1. Advanced Automation & Integrations

- **Communication Webhooks (Slack/Teams):** Automated chat notifications. The system automatically pings the next required approver via chat DM when a release arrives in their queue (e.g., "Hi Release Manager, DevOps has signed off. Your final Stage 5 approval is required.").

2. Intelligent Conflict Resolution & Analytics

- **Automated Resource Recommendation Engine:** If a desired time slot is unavailable due to a shared resource conflict, the platform will suggest alternative open windows or recommend an available backup DevOps engineer.
- **Enterprise Bottleneck Dashboard:** A reporting metrics interface tailored for the Release Manager to track which bottleneck stages (QA, EM, Product, DevOps) hold up approvals the longest over time.

3. Advanced Governance & Document Processing

- **AI-Powered Document Validation:** An integrated LLM parser will scan uploaded SRE Handover Checklists in real-time to verify that the file contains actual rollback code blocks, valid health check URLs, and active contact numbers, automatically rejecting blank documents.